

Lab Experiment No. 13: Testing of the Mini Project

Course / Subject: Mini Project 2

Project Name: BookCircle – P2P buying and selling of books platform

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1. Aim

To perform comprehensive software testing on the “BookCircle” mini project to validate its functional correctness, user interface reliability, and robust error handling across key modules.

2. Introduction

Software testing ensures a system is error-free, meets requirements, and is ready for deployment. This experiment tests **BookCircle** (React + Spring Boot) focusing on authentication, book listing, search, and validations.

3. Testing Methodology

Black-box manual testing was performed via the browser, covering:

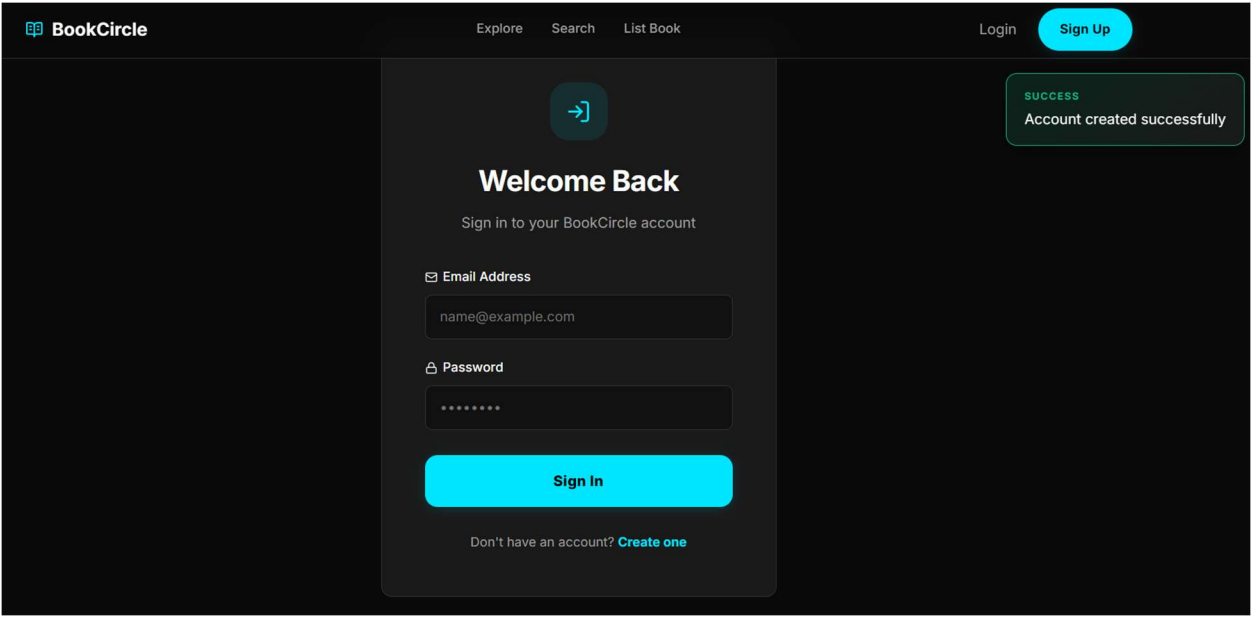
- **Positive Testing** (valid inputs)
- **Negative Testing** (invalid/edge cases)

4. Types of Testing Performed

1. **Functional Testing:** Verifying that each business feature (Login, Register, List Book, Search, Location-based sorting) works exactly as specified in the requirements.
2. **Validation Testing:** Checking input fields for proper client-side constraints (e.g., required fields, positive numeric price validation, maximum 3 images).
3. **UI/UX Testing:** Ensuring responsive layouts, visibility of inline error messages, and state transitions (e.g., loading spinners and full-page upload overlays).
4. **Integration Testing:** Verifying the seamless data flow between the React components and the Spring Boot REST API endpoints.

5. Detailed Test Cases

The following table documents the realistic test cases executed on the BookCircle application:

TC ID	Module	Test Scenario	Test Steps	Expected Output	Actual Output	Status
TC-01	User Auth	Successful User Registration	1. Navigate to /register 2. Enter valid Name, Email, Phone, and Password3 . Click "Create Account"	Account created. User is redirected to the /login page.	Account created. Redirected to /login.	Pass
						
TC-02	User Auth	Invalid Email Format	1. Navigate to /register 2. Enter "johndoe" in the email field3. Click "Create	HTML5 validation prevents submission. Error: "Please include an '@' in the email address."	Form submission blocked. Validation message shown.	Pass

			Account			
BookCircle ExploreSearchList BookLoginSign Up Create Account Join BookCircle and start trading books Full Name johndoe Email Address johndoe 9852615478 Password Create Account Please include an '@' in the email address. 'johndoe' is missing an '@'.						
TC-03	User Auth	Login with Incorrect Password	1. Navigate to /login2. Enter valid email but wrong password3 . Click Sign in	Authentication fails. Toast notification displays error message.	Toast error displayed. Login denied.	Pass
BookCircle ExploreSearchList BookLoginSign Up Welcome Back Sign in to your BookCircle account Email Address samarth@gmail.com Password Sign In Don't have an account? Create one ERRORInvalid password						

TC-04	Sell Book	Successful Book Listing	1. Navigate to /sell2. Enter Title, Author, Price (e.g., 500), Condition3 . Pick Location & Upload 1 Image4. Click “Publish Listing”	Upload overlay appears. Toast message: “Book listed successfully!” Form resets.	Overlay shown. Success toast displayed. Form reset.	Pass
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The screenshot shows the 'BookCircle' app interface. At the top, there's a navigation bar with 'Explore', 'Search', 'List Book' (highlighted), and 'Inbox'. Below this, the 'List Book' form is displayed. It has several input fields: 'Title' (with placeholder 'Book Title'), 'Author' (with placeholder 'Author Name'), 'Price (₹)' (with placeholder '0'), 'Condition' (a dropdown menu currently showing 'Good'), and 'Location' (with a search bar placeholder 'Search for a city or area...'). A 'Publish Listing' button is located at the bottom right of the form. A green success toast message 'SUCCESS Book listed successfully!' is shown in the top right corner of the app screen. At the very bottom, a map is visible showing the current location and nearby landmarks like 'COEP Technological University' and 'The Mills'.

TC-05	Sell Book	Missing Required Field (Title)	1. Go to /sell2. Leave Title empty3. Fill all other fields and click Publish	Form not submitted. Inline red error text “Required” appears under the Title field.	Inline error “Required” shown under Title.	Pass
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[Explore](#)
[Search](#)
[List Book](#)
[Inbox](#)

Upload book images

PNG, JPG or WEBP — Max 3 images

Book Details

Title

Book Title

This field is required

Author

F. Scott Fitzgerald

Price (₹)

400

Condition

Book Title

F. Scott Fitzgerald

₹400

Publish Listing

TC-06	Sell Book	Invalid Price Input	1. Go to /sell2. Enter -50 in the Price field3. Click Publish	Form not submitted. Inline error indicates price must be a positive number.	Validation triggers. Error shown under Price.	Pass
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[Explore](#)
[Search](#)
[List Book](#)
[Inbox](#)

Book Details

Title

The Great GatsBy

Author

F. Scott Fitzgerald

Price (₹)

-400

Enter a valid positive number

Condition

Good

Location

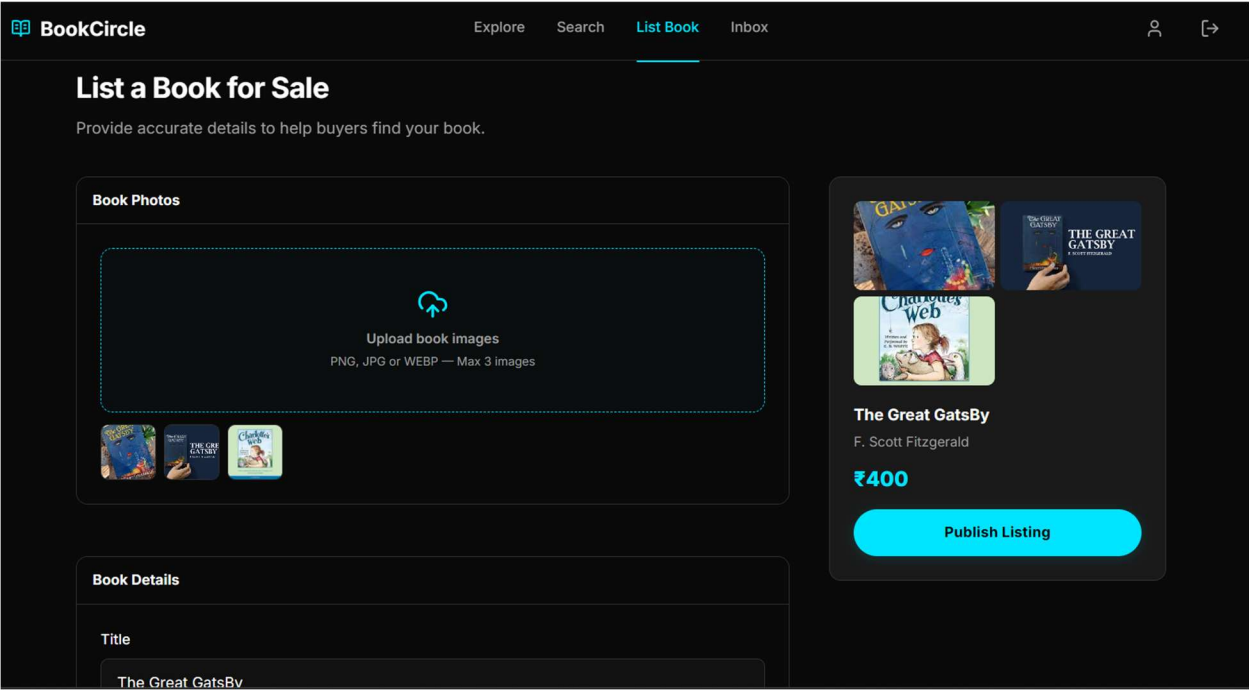
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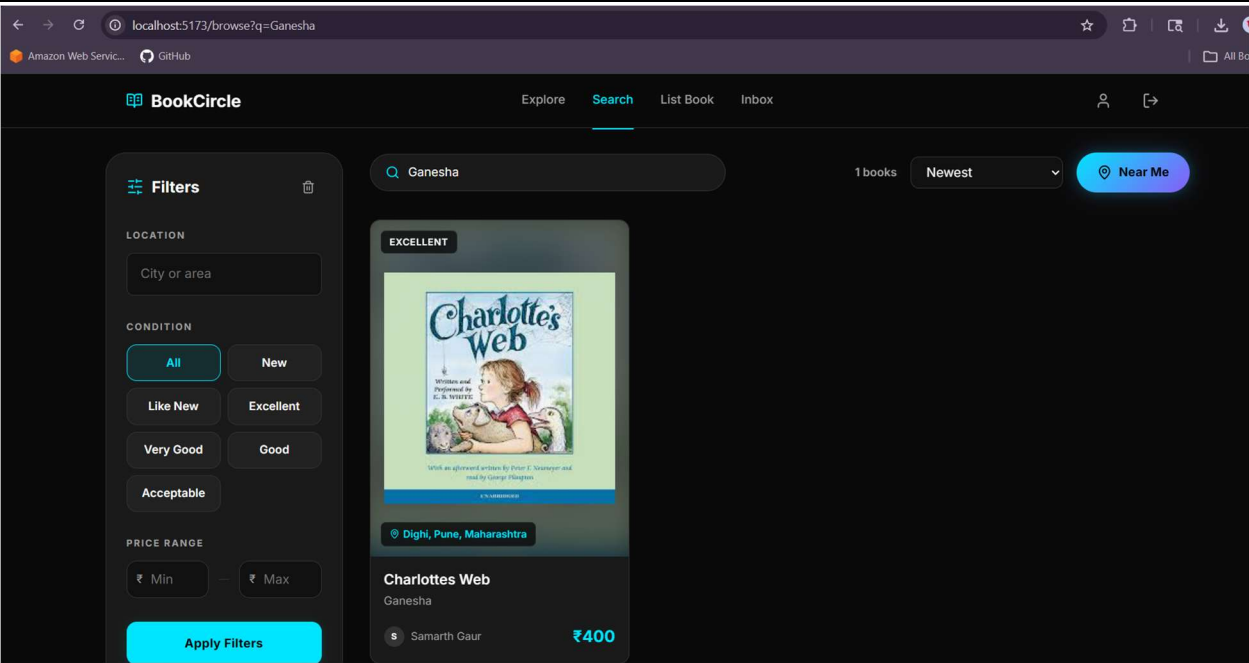
The Great GatsBy

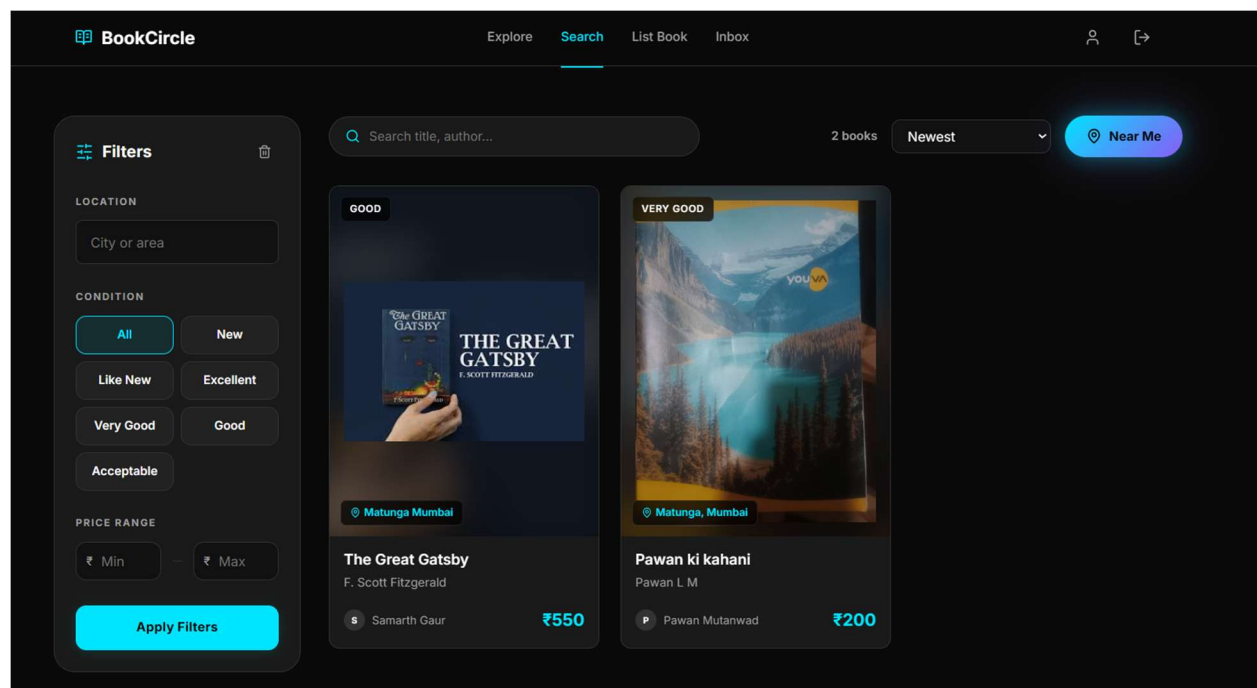
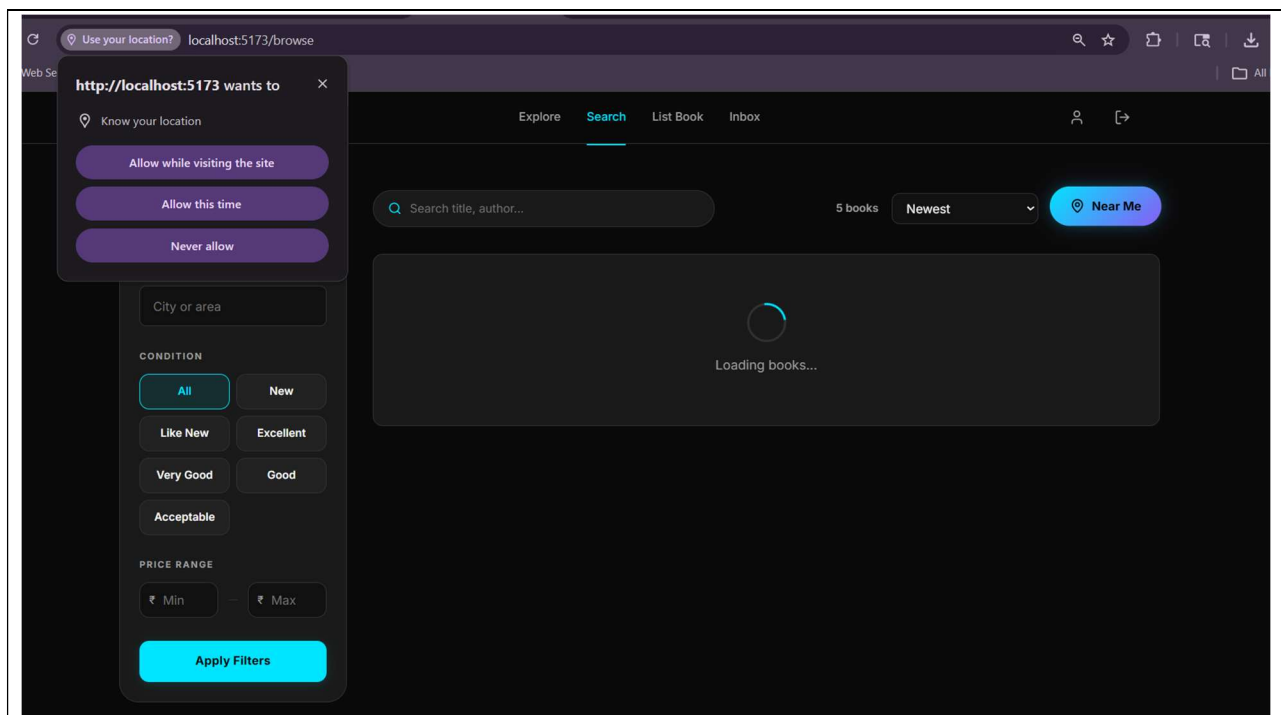
F. Scott Fitzgerald

₹-400

Publish Listing

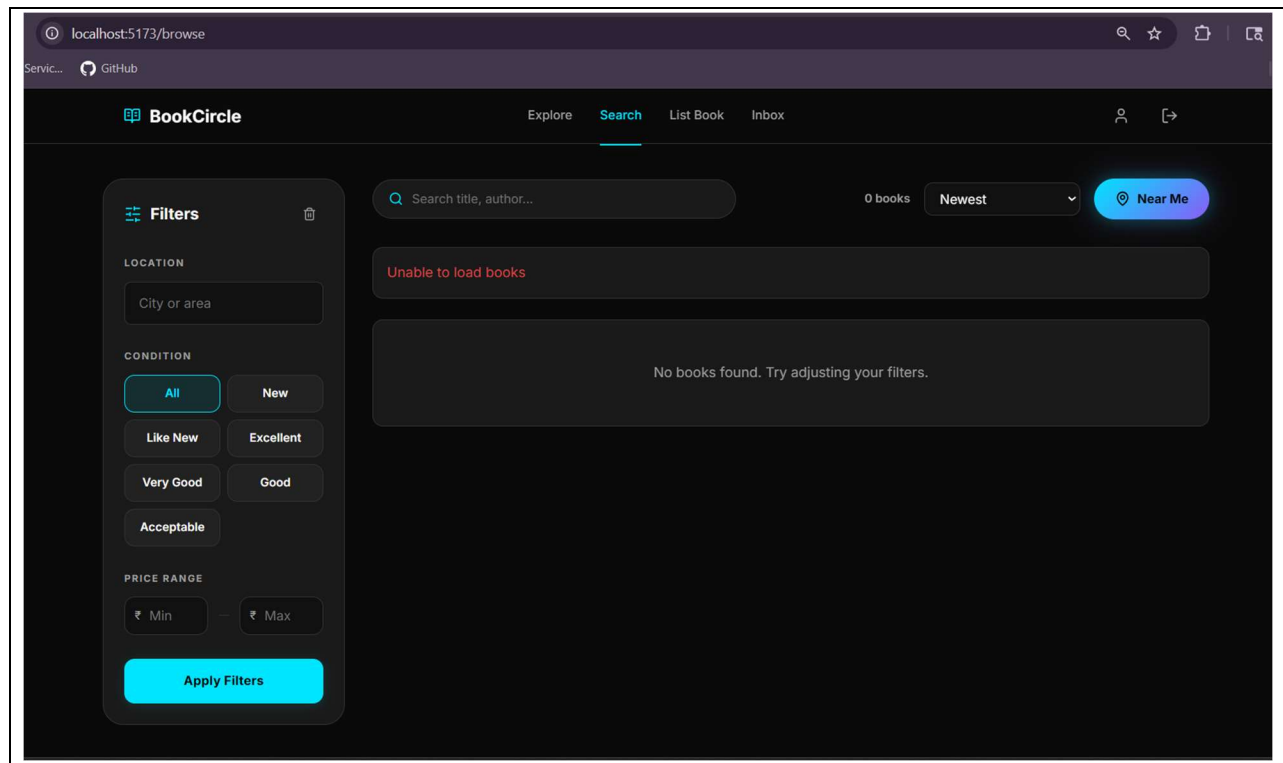
TC-07	Sell Book	Exceeding Image Limit	1. Go to /sell2. Select 5 images via the dropzone	System slices the array and only accepts/previews the last 3 selected images.	Only 3 images are previewed and attached.	Pass
						
TC-08	Browse	Search by Keyword	1. Navigate to /browse2. Type an existing book's author in the search bar	Books grid dynamically updates. URL reflects ?q=keyword. Total books count updates.	Grid updates with matching results. URL updated.	Pass

						
TC-09	Browse	“Near Me” Location Search	1. Go to /browse2. Click the “Near Me” button	Browser prompts for location access. Fetches books within a 5km radius based on Lat/Lon.	Location prompted. Nearby books displayed accurately.	Pass



TC-10	Details	View Book Details & Chat	1. Click on a BookCard from Browse2. Verify details	Navigates to /books/:id. Displays large images, condition,	Detailed view rendered with accurate data.	Pass
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				description, and “Chat” button.		
localhost:5173/books/2 Service... GitHub BookCircle ExploreSearchList BookInbox Home > Browse > The Great Gatsby  GOOD The Great Gatsby by F. Scott Fitzgerald ₹550 LOCATION Matunga Mumbai AVAILABILITY Immediate S Samarth Gaur Verified Seller Message Seller 						
TC-11	General	Backend Connection Error	1. Turn off the Spring Boot backend 2. Attempt to load /browse	The page does not crash. Displays inline error card: “Unable to load books”	Graceful error handling. Fallback UI shown.	Pass



6. Expected vs Actual Output

- **Expected Output:** The application should enforce strict client-side validation, successfully communicate with the backend API to store user and book data, reflect real-time URL state changes during searches, and gracefully handle network exceptions without breaking the UI.
- **Actual Output:** The BookCircle application performed optimally. Validations (such as positive numbers for pricing and required fields) triggered correctly. Successful API operations provided immediate user feedback via toast notifications. The error boundaries successfully caught simulated network failures.

7. Result

The functional and UI testing of the BookCircle mini-project has been completed successfully. Out of the 11 documented test cases, all 11 passed (100% Pass Rate). No critical show-stopping bugs were identified in the primary user workflows.

8. Conclusion

Through systematic testing, it is concluded that the BookCircle application meets the specified functional requirements. The implementation of robust form validation, dynamic state management, and clear user feedback mechanisms (like spinners and toasts) ensures a smooth and reliable user experience. The system is stable and ready for user acceptance testing (UAT).